

How can I tell if my model fits the data in logistic regression?

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 **Partha Talukder** · Follow
M.Sc in Statistics (academic discipline), University of Kalyani (Graduated 2012) · Updated 4y

Originally Answered: How can I tell if my model fits the data in Logistic Regression, the way we use R squared value to determine the goodness of fit in Linear Regression?
I am assuming Logistic regression mentioned here refers to the binary one.

Earlier Mr. Arun Kumar already mentioned one can calculate the **Confusion Matrix** using the actual Labels and the predicted probability from the model.

One needs change the predicted probability threshold (default = 0.5) after scoring on the training data, then test the same with prediction from in-time-validation (from random split), usually the metrics that are looked upon are **Accuracy, Precision, Recall, Specificity,F-measure** etc. and fix the threshold which gives better result on in-time-validation, then go on the check... [\(more\)](#)

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Related **What is logistic regression?**

Hi:

I wrote an article sometime back, hope this is useful.

Introduction to Logistic Regression**Overview-**

Logistic regression is generally used where the dependent variable is Binary or Dichotomous. That means the dependent variable can take only two possible values such as "Yes or No", "Default or No Default", "Living or Dead", "Responder or Non Responder", "Yes or No" etc. Independent factors or variables can be categorical or numerical variables.

Please note that even though logistic (logit) regression is frequently used for binary variables (2 classes), it can be used for categorical dependent v... [\(more\)](#)

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To assess how well your logistic regression model fits the data, you can use several methods and metrics. Here are some key approaches:

1. Confusion Matrix

- The confusion matrix provides a summary of prediction results on a classification problem. It shows the counts of true positives, true negatives, false positives, and false negatives. This helps you understand how well the model is performing.

2. Accuracy, Precision, Recall, and F1 Score

- Accuracy:** The proportion of true results (both true positives and true negatives) among the total cases.
- Precision:** The ratio of true positives to the total pre

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Logistic regression is applied only when the outcome variable is binary. For example, Customer will churn or not, prospect is eligible for loan or not, customer is eligible for discount or not, there is anomaly activity or not in the system, transaction is fraudulent or not etc.

If you mean that how to judge whether logistic regression model is good fit or not, you need to compute Confusion Matrix.

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Absolutely not

Definitely yes

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Related **In a logistic regression model (SAS), if the data has more than 30 variables, how do I find out which variable can fit the model?**Adding to excellent answer by [Paritosh Gupta](#) :

- Fit the plain vanilla model using all and then look at absolute value of the t-statistics for each variable. This is one way to understand variable importances. This helps when bivariate and WOE methods lead to confounding.
- Look at some statistics like goodness of fit, pseudo R-squared. See addition/deletion of which variables leads to better value.
- Look whether principal components provides you with a better set of latent features. (might not happen always)
- Look for interactions(business knowledge helps) among the variables, they might provide signif

... [\(more\)](#)

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Related **What is logistic regression in layman's terms?**

I really like answering "laymen's terms" questions. Though it takes more time to answer, I think it is worth my time as I sometimes understand concepts more clearly when I am explaining it at a high school level.

I'll try to make this article as non-technical as possible by not using any complex equations, which is a challenge for a math junkie such as myself. But rest assured, this won't be a one-liner.

You may have heard about logistic regression from a blog, newspaper article etc. You'll only understand what it is when you understand what it can solve.

Problem:
Let us examine a simple and a ve... [\(more\)](#)

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Related **In a logistic regression model (SAS), if the data has more than 30 variables, how do I find out which variable can fit the model?**

There are multiple ways for variable reduction techniques-

- You can do bivariate analysis to check the predictiveness of variable
- You can check the Information value of variable using WOE transformation
- Most easily is to do run a PROC VARCLUS to do clustering and find the most important variables from each clusters

Hope this help.

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Related **How would you evaluate a logistic regression model?**A logistic regression model can be evaluated by looking at the confusion matrix. The accuracy, sensitivity and specificity can be good indicators of your model and what you want to do with your model - concentrate more on true positives or false negatives. Also, you can use precision and recall and use the f1 to evaluate your model. Precision and Recall basically evaluate your model based on how many true positives you get out of total positives and how many truepositives you get out of the total number of positives you predicted. Ofcourse, there is a trade off between Precision and Recall.If ... [\(more\)](#)

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Related **How would you evaluate a logistic regression model?**

The evaluation of a logistic regression model is not quite as straight forward as the linear regression model. In a linear regression model we can simply calculate the value of R-square which acts as a very good measure of the credibility of the model.

In case of logistic regression we do not have such a straight-forward measure. In case of logistic regression model we can use AIC, BIC.

Akaike Information Criteria

We can use the AIC when we are comparing many models and we have to select one of them.

 $AIC=2k + 2\log(L)$ where L denotes the likelihood of the model k is the number of the parameters.It ... [\(more\)](#)

 **Phillip Adkins** · Follow

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or hierarchical models on?

Rather than find one for you, I'll tell you how I'd find it.

Note that logistic regression minimizes a "log loss" or "cross entropy error". Make sure you know what that loss function looks like when written in summation notation.

Now - go to past competitions, and look at the "evaluation" page. They'll show the cost function that you're to optimize there.

Related questions

- How would you evaluate a logistic regression model?
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Related **How is predictive modeling used in logistic regression?**
Logistic Regression — Detailed Overview

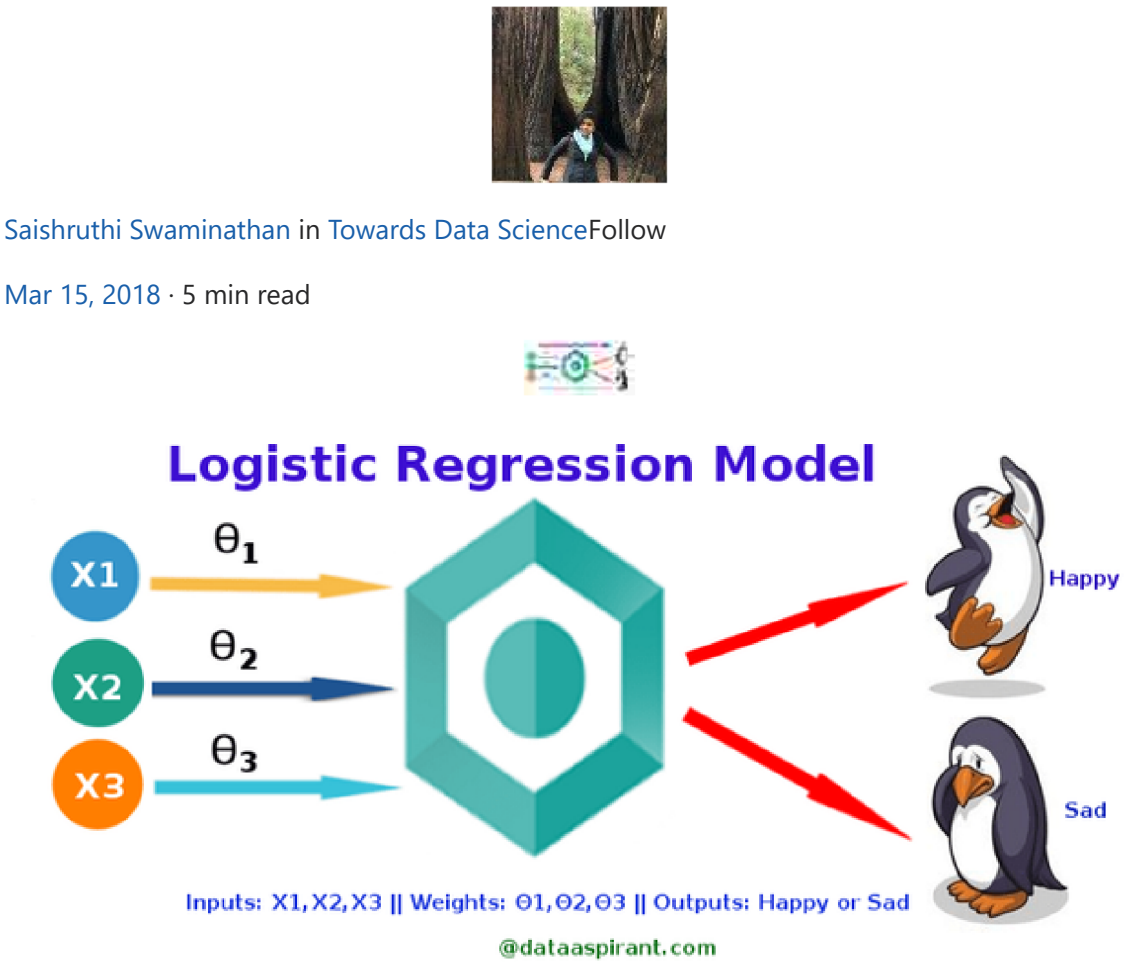


Figure 1: Logistic Regression Model (Source:<http://dataaspirant.com/2017/03/02/how-logistic-regression-model-works/>)

Logistic Regression was used in the biological sciences in early twentieth century. It was then used in many social science applications.

Logistic Regression is used when the dependent variable(target) is categorical.

For example,

- To predict whether an email is spam (1) or (0)
- Whether the tumor is malignant (1) or not (0)

Consider a scenario where we need to classify whe... (more)

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Related **How do I interpret the coefficients for logistic regression?**
The logistic regression is given by

$$\pi_i = P(Y_i = 1|X_i = x_i) = \frac{\exp(\beta_0 + \beta_1 x_i)}{1 + \exp(\beta_0 + \beta_1 x_i)} \tag{1}$$

By algebraic manipulation, the logistic regression equation can be written in terms of an odds ratio for success.

$$\begin{aligned} \text{logit}(\pi_i) &= \log\left(\frac{\pi_i}{1 - \pi_i}\right) \\ &= \beta_0 + \beta_1 x_i \\ &= \beta_0 + \beta_1 x_{i1} + \dots + \beta_k x_{ik} \end{aligned} \tag{2}$$

We can interpret the logistic regression in three ways

Based on the coefficient sign: The equation 1 shows the relation between the probabilities of class 1 and the ... (more)

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Related **How is predictive modeling used in logistic regression?**
I think the question is better phrased: "How is logistic regression used in predictive modeling?"

To answer that question, we first need to look at what logistic regression accomplishes. The first and foremost result of a logistic regression is that for a set of independent variables it can give you a binary 0 or 1 for a single dependent variable. If you are able to get a good fit/model, then you can plug in the independent variable values for a new observation and predict if the dependent value will be 0 or 1.

For example, let's say you have a set of data that has independent variables age, ge... (more)

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Related **Why does logistic regression not converge when the data is completely separable?**

The question is wrong. The key concept here is "converge".

How is "converge" defined? In machine learning, "converge" means that an algorithm can reach the ideal solution within a predefined distance. Often it means that we can reduce the error to less than a predefined threshold. Otherwise, we call the algorithm "not converge". For example, for a neural network model, if the learning rate is too big, the algorithm may not be able to reduce the training error to low enough.

Having multiple solutions or having infinity many solutions don't mean that the algorithm can not converge as long as the a... (more)

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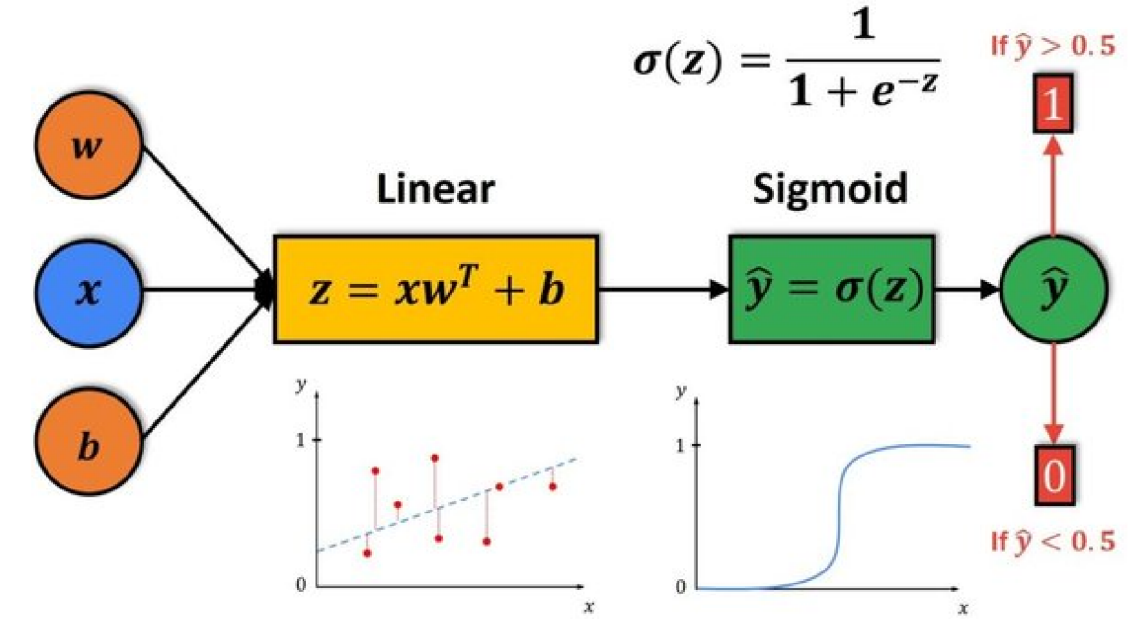
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Related **Can we use logistic regression for continuous variables?**

On a continuous variable, Yes. **To predict** a continuous variable, No.

Logistic regression is designed for classification and it achieves that by squashing the output space into a binary one.



This is your regular linear regression function : $f(x) = wx + b$

When you feed it to a Sigmoid function $f(x) = \frac{1}{1+e^{-x}}$ you reduce the output space into very restricted one $[0, 1]$

The final result will be even more restricted, $(0, 1)$ as it will be matched to 1 if the output is greater than 0.5 and zero otherwise... (more)

1 answer collapsed (Why?)

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How do I combine two logistic regression models trained on different sets of data?

How do I do logistic mixed regression with this data?

How do I manage a logistics regression model?

How would you create a logistic regression model?

Is logistic regression just linear regression on a log scale?

What is the effect of not having an intercept term in logistic regression models?

Why is logistic regression considered a linear model?

If the target variable is not available directly in a dataset for logistic regression, how do we compute it?